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Electoral geography: From mapping votes to representing power

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Abstract

In some ways, electoral geography has never been more popular. From the detailed, online maps of the Brexit vote to discussions of the electoral college versus the popular vote in the 2016 Trump–Clinton U.S. presidential contest, the relationships among geography, voting, and political power have seldom been more visible. The popularity of electoral geography in social and news media, however, does not necessarily reflect its presence in scholarly discussions, and indeed, in some ways, the former has replaced the latter. Digital technology and the burgeoning availability of electronic data mean that it is easier than ever to create maps of votes, often in near real time. Yet the academic field of electoral geography encompasses more than just mapping votes, including the study of election campaigns, political parties, electoral systems, and gerrymandering. The 3 major approaches are the geography of voting (mapping and visualizing votes), geographic influences on voting (the effect of place on political preferences and behavior), and the geography of representation (the analysis of electoral systems). Indeed, the structure of the electoral system, including gerrymandering, is often the key to understanding how political and racial/ethnic minorities can (or cannot) wield power and influence. This article examines each approach after a brief review of the historical origins of the subfield.

1 | INTRODUCTION

The success of the British referendum to leave the European Union (“Brexit”) and the victory of Donald Trump in the U.S. presidential election in 2016 raised doubts in the popular media about the accuracy and efficacy of political polling. In the popular narrative, pollsters—through incompetence, hubris, or political bias—made inaccurate predictions. Scholarly analysis suggests that the results of both contests were within the margin of error of most polls, and that these “unexpected” outcomes were more a product of media campaign coverage and misunderstanding

the nature of statistical predictions (e.g., Campbell et al., 2017; Searles, Ginn, & Nickens, 2016). Nonetheless, these two cases served as a useful reminder that polling data are distinct from election results, and that the former are not always a perfect predictor of the latter. They also show the extent to which we now rely on polls as guides to public opinion and political preference. Yet for much of the 20th century, scholars used election results rather than polling data to make inferences about the causes of political preferences. This kind of inferential electoral geography, however, has now been largely eclipsed by other methods and approaches.

Among other factors, the rise of high-quality polling data in the mid-20th century and the concomitant capacity to tie the political preferences of individuals to their specific socio-demographic characteristics (age, gender, education, etc.) led to a decline in the popularity of electoral geography analysis. Analysts can use polling data to assess causal effects directly, rather than relying on the aggregated data of election returns to infer causation (King, 1996). Nonetheless, political geographers still analyze aggregate election data to infer causal relations, particularly in situations where high-quality polling data is not available (e.g., Clem & Craumer, 2004; 2005).

More recently, inexpensive, powerful digital technologies have made cartographic representations of electoral data common in news and social media. Election coverage on television, for example, now commonly features a variety of maps, including highly detailed interactive ones that are updated in near-real time (Field & Dorling, 2016; Miller, 2016). Web-based cartographic tools also permit non-specialists to generate and share political and election-related maps (Crampton, 2009; Davisson, 2011). Election mapping, in a sense, has become democratized (Shin, 2009).

In short, in the mid-20th century, poll-based data begin to challenge the utility of electoral geography for causal analysis and inference, and in the late 20th century, accessible digital technology challenged academic geography's dominance of political cartography. Nonetheless, contemporary electoral geography continues to make valuable scholarly contributions (Warf & Leib, 2011), particularly as related to the analysis of spatial patterns of voting, the influence of geography on voting, and—more broadly—on the geography of political representation.¹

2 | THE ORIGINS OF ELECTORAL GEOGRAPHY

The first work of electoral geography is generally considered to be André Siegfried's (1913) *Tableau politique de la France de l'ouest sous la troisième république*, although in the Anglo-American tradition, Prescott (1959) and Rowley (1970) identify Krebheil's (1916) analysis of British Parliamentary elections as the earliest work. Schulten (2012) suggests, however, that the map of the 1880 presidential vote in *Scribner's Statistical Atlas of the United States* published in 1883 could be considered the first modern election map (Figure 1).

Both Siegfried and Krebheil ask similar questions and employ similar methods: What is the spatial pattern of a vote, and what explains such patterns? As such, both ultimately sought to make a causal argument about the source of political preference and voting behavior. In essence, they compared maps of election returns with physical or social factors, and interpreted similarities in spatial patterns as evidence of causality. Krebheil identified variations in occupation as the key factor, while Siegfried focused on social variation caused by different physical environments (geology). (Siegfried was not an environmental determinist, however, and considered physical factors as a proxy for economic and social patterns.) Subsequently, in the United States, John K. Wright (1932a, 1932b) introduced the mapping of legislative (Congressional) votes, in addition to the mapping of election returns. This was an important conceptual innovation because it focused concern on political relationships per se, rather than using election results as a proxy for political preferences.

Early electoral analyses were undertaken before the advent of modern survey methods and before the invention of digital technology powerful enough for systematic spatial analysis (but for an early quantitative analysis, see Warntz, 1955). This meant that election returns were the best—if not only—source of information about variations in public opinion and political preference. Without systematic data on individual responses, however, causality could only be inferred by spatial coincidence. Hence, Krebheil interpreted his findings (that industrial and less wealthy agricultural regions tended to support the Liberal Party while wealthier agricultural areas were Conservative bastions) to

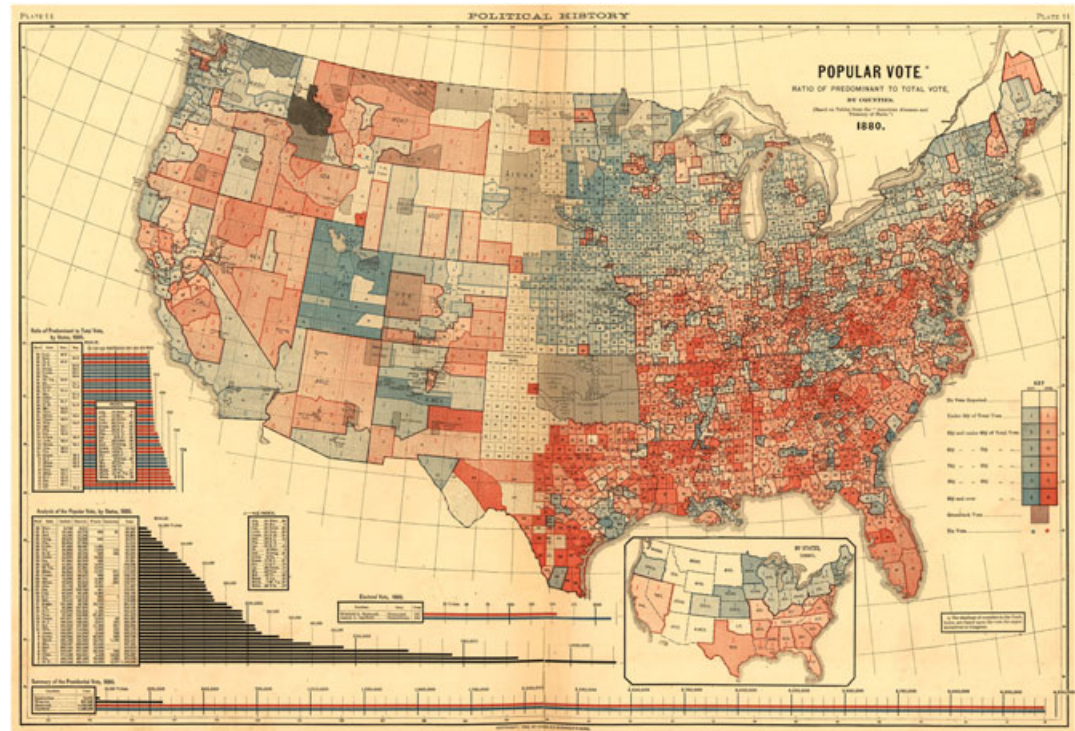


FIGURE 1 Map of the 1880 presidential popular vote by county in *Scribner's Statistical Atlas of the United States* (1883). Source: Library of Congress Geography and Map Division Washington, D.C. USA. Digital ID: <http://hdl.loc.gov/loc.gmd/g3701gm.gct00009>

mean that there was a causal relationship between occupation and political preference. This relationship could be affected, however, by party positions on specific political issues, e.g., support for Home Rule in Ireland explained the higher-than-expected support for the Liberal Party in southern Ireland. Such analyses were not statistical, however, and relied primarily on the visual inspection of maps. They also suffer from the ecological fallacy, imputing individual characteristics (and causal mechanisms) from aggregate data. Regardless of such analytic limitations, electoral mapping during the late 19th and early 20th century was used by parties and governments for many ends, including nation-building and partisan competition (de Jong & Kaal, 2017; Retallack, 2016).²

While techniques of electoral analysis have grown more sophisticated, the basic approaches, questions, and limitations in much contemporary electoral geography remain similar to these early works.

3 | THE GEOGRAPHY OF VOTING

Until the advent of sufficiently powerful and inexpensive computer technology (roughly the early 1990s), the task of mapping election returns remained relatively difficult, giving academic geographers (along with major news organizations and government agencies such as the Census Bureau) close to a monopoly on electoral cartography. Work in the 1930s, such as Wright (1932a, 1932b), showed maps of U.S. presidential elections, with some inferences about the explanation of persistent patterns. The main thrust of such work, however, was the visualization of election results and the identification of spatial patterns.

With some exceptions, most research in electoral geography during the postwar period was by French scholars (Morazé, 1947; Siegfried, 1949), but the geography of voting perspective enjoyed a significant revival among

Anglo-American geographers in the early 1980s with works emphasizing the persistence of regional cleavages in American elections (Archer, 1988; Archer, Murauskas, Shelley, White, & Taylor, 1985; Archer & Taylor, 1981; Martis, 1988; Shelley, Archer, & White, 1984). Indeed, Archer and Shelley (1986) co-authored a "Resource Publication" of the Association of American Geographers that declared, "the task of the political geographer is the identification and interpretation of the sectional manifestations of political issues." This approach, emphasizing the identification of patterns of voting through cartography and spatial analysis, continues with the production of political atlases (Archer, 2006; Archer et al., 2014; Brunn, Lavin, & Archer, 2011; Leonard & Natkiel, 1987; Martis & Rowles, 1982; McRobie, 1989; Romero Ballivián & Basset, 2007; Sallnow, John, & Webber, 1982; Salmon, 2001).

By the 1980s, however, academic geographers were losing their monopoly on electoral mapping as high-quality cartography became more common outside the scholarly community. Archer's (1985) article on the 1984 U.S. election, for example, reproduces maps from *Time Magazine*. As information and **geographic information systems** technology improved through the 1990s, it became even easier for media organizations to create maps showing election returns. The ubiquitous "red-blue" map of the 2000 U.S. presidential election cemented the spread of electoral cartography as well as the association of color with the two major U.S. parties (Shin, 2009; Figure 2).

The popularity of the "red-blue" map and the distortions inherent in such a representation sparked responses in academia, news media, and social media (Herb et al., 2009). Two kinds of "corrections" are particularly common. First, "purple maps" replace the binary red-blue scheme with a graduated color scale that reflects the degree of support for presidential candidates, as in Figure 3. Second, cartograms with areas corresponding to population or votes are used to correct the visual bias caused by large, sparsely populated regions (Figure 4).

Both techniques are well-known in academic cartography, but have been applied by those without specialized training in geography, and are available easily via the Internet, illustrating degree to which disciplinary geographers

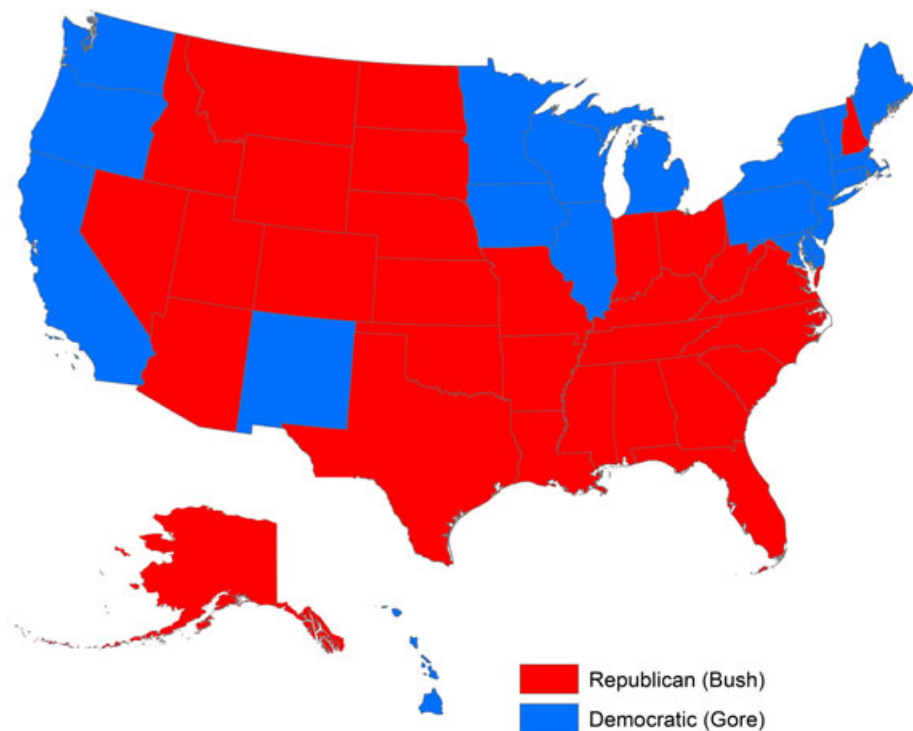


FIGURE 2 Electoral vote winner by state in the 2000 U.S. presidential election: The "Red-Blue Map." (Source: Map by author; data from Federal Election Commission, www.fec.gov/pubrec/fe2000/ptables.xlw.)

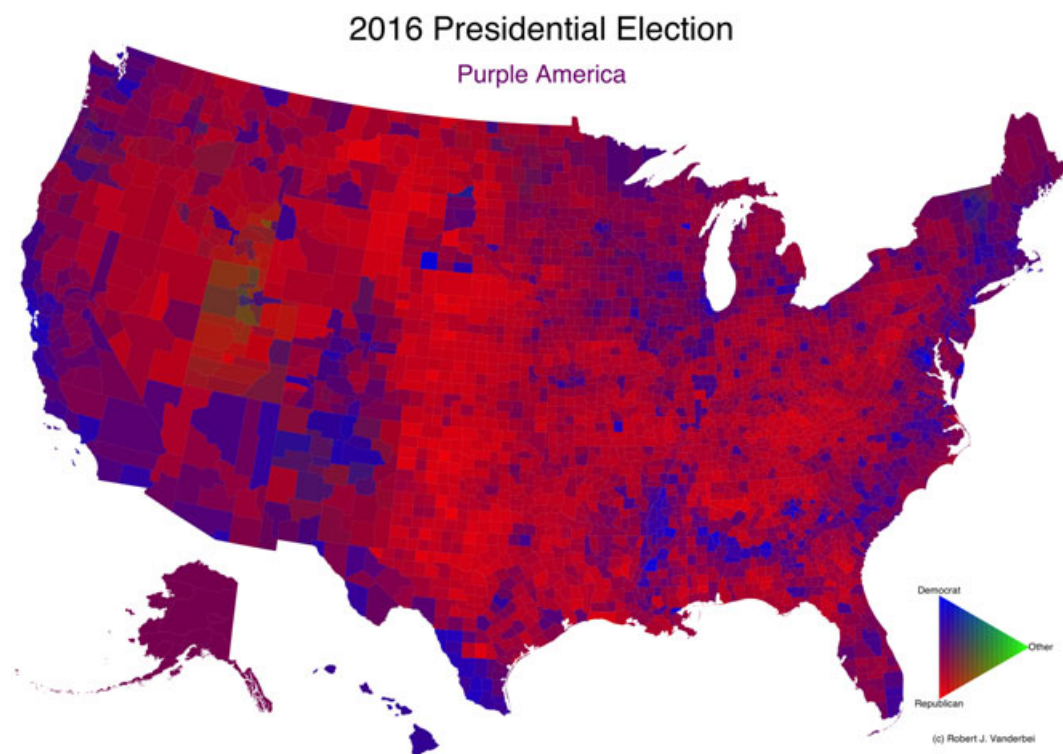


FIGURE 3 Popular votes by county in the 2016 U.S. presidential election with graduated color scale: The “Purple Map.” (Source: Map from Prof. Robert J. Vanderbei, Princeton University, www.princeton.edu/~rvdb/JAVA/election2016/)

no longer hold a monopoly on electoral mapping. At the same time, scholarly cartography continues to grapple with the challenge of representing electoral data in clear and effective ways (e.g., Ondrejka, 2016).

While work like Archer's, and more recent ones in the same vein (e.g., Clem, 2006; Heppen, 2003; Heppen & Mesyanzhinov, 2003; Kinsella, McTague, & Raleigh, 2015; Morrill, Knopp, & Brown, 2007, 2011; Walks, 2005; Walks, 2010; Webster, Chapman, & Leib, 2010; Webster & Quinton, 2010) employ sophisticated statistical analyses to identify spatial patterns, they remain primarily concerned with *where* votes are cast. The focus on the spatial pattern of voting in and of itself has obvious limitations. In general, spatial patterns in voting can be attributed to two factors: compositional and contextual effects. Compositional effects refer to the patterns produced because the composition and characteristics of the population (or rather the electorate) varies over space. Some variation in voting patterns results from different kinds of people living in different places. In contrast, contextual effects are where voters' social environments (broadly conceived) and/or social networks shape their political preferences independently from their individual characteristics. Some variation in voting patterns is thus due to differences in social environments, local traditions, and place-bound political organizations.

Electoral cartography cannot alone distinguish compositional and contextual effects. To return to Krebheil's work, if a survey reveals that industrial workers favor Liberal Party policies, creating a map showing concentration of support for Liberals in areas where industrial workers live adds little to our understanding of this relationship. A few analyses have attempted to disaggregate the contribution of compositional and contextual effects on the spatial pattern of political preference, but such research requires detailed survey information and is difficult to implement for large areas (Gent, Jansen, & Smits, 2014; Walks, 2006).

Moreover, gender—often a major determinant of political preference—tends to be spatially uniform and thus “invisible” to electoral analysis. With polling and survey data, the influence of such various factors can be assessed

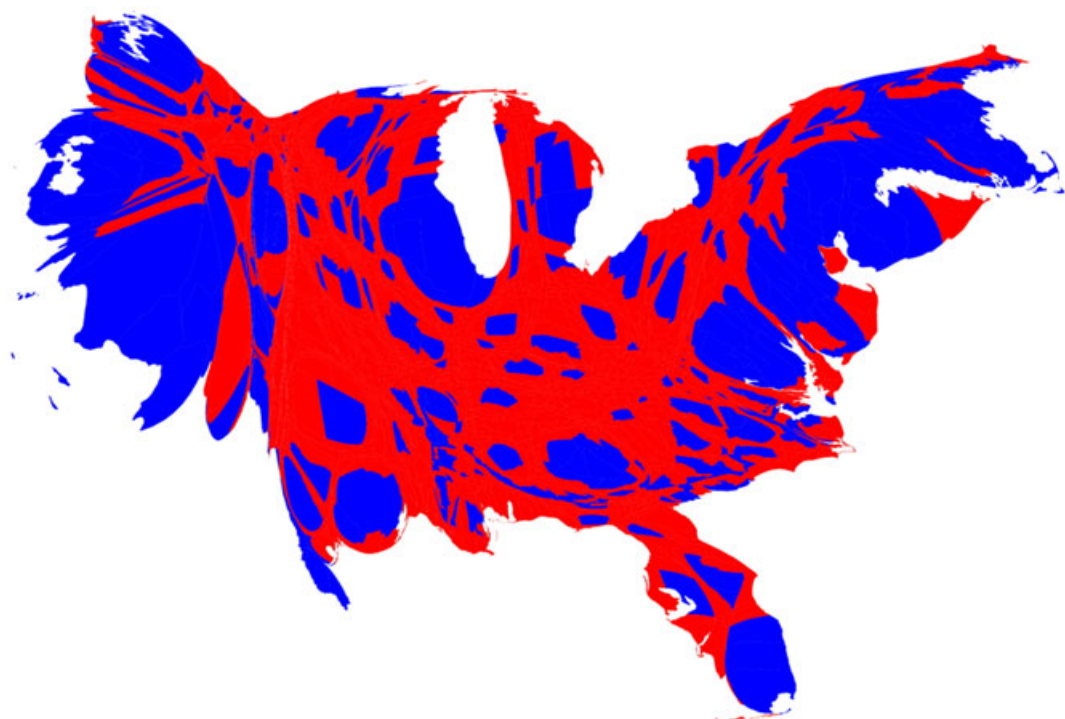


FIGURE 4 Cartogram of popular vote winner by county in the 2016 U.S. presidential election (Source: Map from Professor Mark Newman, University of Michigan, www-personal.umich.edu/~mejn/election/2016.)

directly, so drawing causal inference from spatial patterns becomes a poor alternative. Consequently, scholarly attention to individual causality has shifted largely to the analysis of survey data in political science. Although some geographers use polling data, the practice is relatively limited among electoral geographers (notable exceptions include O'Loughlin, Vladimir, & Olga, 1997, Pattie & Johnston, 2000, and Walks, 2007).

Moving beyond the compositional effects shown by the geography of voting has been one of the major challenges for electoral geography. As Agnew writes, " 'Mapping politics' can offer more than cartographic illustrations that decorate more compelling aspatial accounts of electoral geography, but only if we work harder at understanding the roles of context and showing in what ways it counts" (Agnew, 1996, 144).

4 | GEOGRAPHIC INFLUENCES ON VOTING

A second major theme is geographic influences on voting. Interest in this issue appears in the earliest works in electoral geography: What causes people to vote differently in different areas? As geography embraced statistical and quantitative techniques in the 1960s, it became possible to answer that question in a more systematic fashion (e.g., Cox, 1969). Ron Johnston (2005) has repeatedly articulated a contemporary version of this question: Do "similar people vote in the same way wherever they live?" Indeed, the question of whether "place matters" is one of the major points of dispute between political science and political geography (Johnston & Pattie, 2006, 40–43). Political scientists such as Gary King (1996) argue that political behavior and preference can be explained by purely individual factors, and that unexplained spatial variation is merely the consequence of imperfect knowledge about individual voters, i.e., compositional effects. Agnew (1996, 2002), Johnston (1986), and Charles Pattie (Pattie & Johnston, 1997; Johnston & Pattie, 1998; Pattie & Johnston, 2000; the latter two typically working together) have carried on

a spirited debate with this position, arguing for the contextual approach and for the existence of the “neighborhood effect.”

The contextual perspective asserts that political preferences and behavior cannot be understood or explained by looking simply at individual characteristics (income, occupation, age, etc.) but depend to some degree on place and location (Flint, 1996; Johnson, Phillips Shively, & Stein, 2002; O'Loughlin, Flint, & Anselin, 1994; Pattie & Johnston, 1999; Sui & Hugill, 2002; Vilalta y Perdomo, 2004). For example, early work—ironically by political scientist Key (1949)—found that support for candidates was relatively greater in areas close to the candidate's home, an approach and finding that is still replicated in contemporary geographic research (Gimpel, Karnes, McTague, & Pearson-Merkowitz, 2008).

Contextual effects go beyond mere distance, however. Johnston and Pattie (2006) identify at least eight scales (from the household to neighborhood to country) that may constitute the “place” of an individual and that can influence his or her vote. One can imagine, for example, that a growing national economy may encourage a favorable view of the incumbent party among voters in general, but may have little effect if one's spouse is recently unemployed (i.e., at the scale of the household, the economy is doing poorly). Walks (2006) argues that place-specific characteristics are among the factors that explain variations in political preference (e.g., compared to suburbanites, urban dwellers favor expenditures on collective goods like public transportation). Most commonly, however, contextual effects are investigated at the neighborhood scale, hence the term the “neighborhood effect.” Such an effect can result in greater political homogeneity through person-to-person interaction, the adoption of common role models, and through the influence of institutions (common schools, churches, etc.) Fieldhouse and Cutts (2008) show, for example, that the turnout of minority groups increases as the proportion and absolute number of minorities in the neighborhood increases. Alternatively, residents of a neighborhood may compete with each other for resources, or may judge their interests and status relative to their immediate neighbors rather than to a national average. These latter two may produce greater heterogeneity of preferences than otherwise expected. Pattie and Johnston (2000), use data from the 1992 British Election Study to show that voters who spoke together tended to change their party preferences to match each other. (Also see their updated research in Johnston, Wickham-Jones, Pattie, Cutts, & Pemberton, 2016.) Such face-to-face conversations are a consequence of living in the same area, and cannot be reduced to individual-level characteristics. At the same time, the effect of such information sharing has not always been shown to be significant (e.g., Sui & Hugill, 2002; Walks, 2006), so the relative importance of the neighborhood effect remains an active area of research.

Beyond face-to-face interactions, electoral geography has generally focused on political party activities as a contextual force (e.g., Flint, 2001; Rodden, 2010). In their study of Italian elections, for example, Shin and Agnew (2008) and Agnew (1996, 139) identify two contextual effects that are closely tied to party activity and presence. First, the slate of parties may not be the same in every constituency, so that voters in different areas “consider the political options and what they mean in different ways.” Second, parties may have more or less effective presence in different areas, with some having “familiar faces to present, congenial stories to tell, and the means to communicate them both” while others do not.

In a different fashion, spatial analyses of campaign spending also reflect a perspective that asserts that location makes a difference in political preference. The mechanism in this case is slightly different. Political parties may “invest” resources in particular areas, building the kind of long-term presence that may produce local exceptions to national patterns (Forrest, 1997). In electoral systems that use territorial constituencies, such as Canada, the United Kingdom (UK), and the U.S., parties typically vary campaign spending among constituencies and media markets. As Pattie and Johnston (Pattie & Johnston, 2003; Pattie & R., 2009) have shown in the UK, parties typically maximize expenditures in their most competitive constituencies, where their candidate has a relatively even chance of winning or losing. Parties can thus create spatial variation through both long- and short-term activities.

With only a few exceptions (e.g., Smith & Hart, 1955; Webster & Leib, 2001), electoral geography has focused on election and referenda analysis, and have not pursued Wright's investigation of legislative (roll call) votes. This is unfortunate because such analyses can reveal the dynamics of the relationship between representatives and their constituents in a geographically sensitive manner.

5 | THE GEOGRAPHY OF REPRESENTATION

The third major area of electoral geography concerns the analysis of electoral systems. The preponderance of this research is on so-called “first-past-the-post” (FPTP) or plurality systems using territorial constituencies. Typically in such systems, two or more candidates compete for a single seat, and the voting constituency is a contiguous geographic area (“districts” in the United States, “ridings” in Canada, and simply “constituencies” in the UK). Generally, the candidate who obtains a plurality or majority of votes wins the seat, although Australia combines constituency-based representation with forms of proportional representation (Johnston & Forrest, 2009). Some countries, such as New Zealand and Germany, use a combined system where some representatives are elected from territorial constituencies and some by proportional representation. Outside the Anglo-American tradition, however, proportional representation (PR) is far more common (Colomer, 2004, pp. 3–78, Lijphart, 2012). In PR systems, an elector casts a vote for a party rather than a candidate, and parties obtain seats (more or less) in proportion to the number of votes they received.

The reasons for the focus on plurality systems are arguably two-fold. First, the constituency system provides data that easily demonstrate spatial variation. The 338 ridings of the Canadian House of Commons, the 435 Congressional districts of the U.S. House, or 650 seats of the British House of Commons, for example, all provide scholars with multiple simultaneous elections with thousands of candidates, multiple parties, and a wide variety of socioeconomic and geographic contexts. Modern computer and **geographic information systems** technology has increased the availability and detail of such data to an extraordinary degree. These data not only lend themselves easily to spatial analysis, but election outcomes in FPTP systems are highly dependent on the geographic distribution of the vote. Nonetheless, there is some creative work on the geography of representation in PR systems. Recent research on Belgium, for example, analyzes the residential location of representatives elected under a PR system (De Maesschalck, 2011; Put, 2016; Put, Maddens, & Verleden, 2017).

Second, Anglo-American geographers tend to live in countries that use plurality systems. To the extent that scholarly interest and research funding stay close to home, the focus on such systems is hardly surprising. Indeed, Ronald Johnston and his collaborators are so dominant that their analyses of the British system constitute a large plurality of publications in the subfield. De Maesschalck (2011), De Maesschalck and Loopmans (2003), and Put (2016) are exceptions that prove the rule, as they analyze election outcomes in their home country of Belgium, which uses a PR system. Nonetheless, the collection edited by Warf and Leib (2011) includes case studies from Europe and Asia in addition to those of the UK and USA. There is also periodic attention to historical cases (Flint, 2001; O'Loughlin et al., 1994), and elections outside Europe and the Anglo-American core (Clem, 2006; Fox & Lemon, 2000; Hsu, 2009; Martis, Kovacs, Kovacs, & Peter, 1992; Osei-Kwame & Taylor, 1984; Perepechko, Kolossov, & ZumBrunnen, 2007; Vilalta y Perdomo, 2004).

5.1 | Converting votes to seats

The nature of plurality systems means that, counterintuitively, FPTP elections rarely produce proportionality because candidates compete in separate constituencies and can win (in principle) by a single vote. That is, the number of seats a party wins in a legislature may not reflect the total number of votes the party's candidates received nationally. If a party's candidates, for example, come in second place in all districts, the party would not win a single seat even if they obtain a significant fraction of the vote. Figures 5-7 show the relationships between the percentage of seats won and the percentage of the national vote for political parties in the USA, UK, and Canada in the post-World War Two era. The basic relationship between seats and votes follows the so-called cubic law (Kendall & Stuart, 1950), but this relationship depends on both the boundaries of constituencies and the geographic pattern of the vote in each election (Johnston, 2002a, 2002b; Warf, 2008; Johnston & Pattie, 2011). In general terms, plurality systems penalize parties whose support is spread relatively evenly, unless it is high enough to constitute a plurality in many constituencies. A FPTP system can provide opportunities,

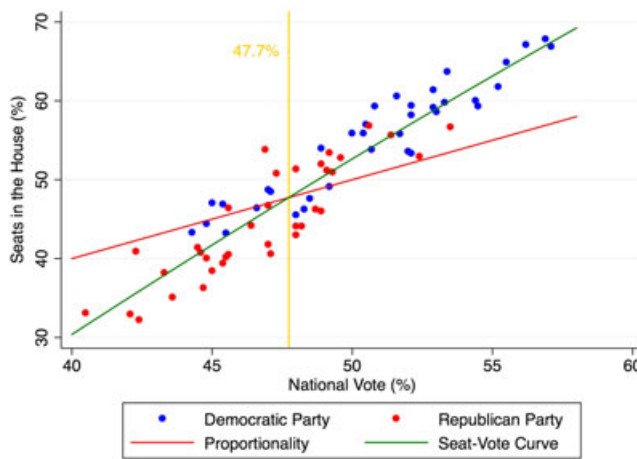


FIGURE 5 Votes and seats by party in the U.S. House of Representatives: 1946–2016 (Source: Brookings Institution, Vital Statistics on Congress, chapter 2. www.brookings.edu/multi-chapter-report/vital-statistics-on-congress/)

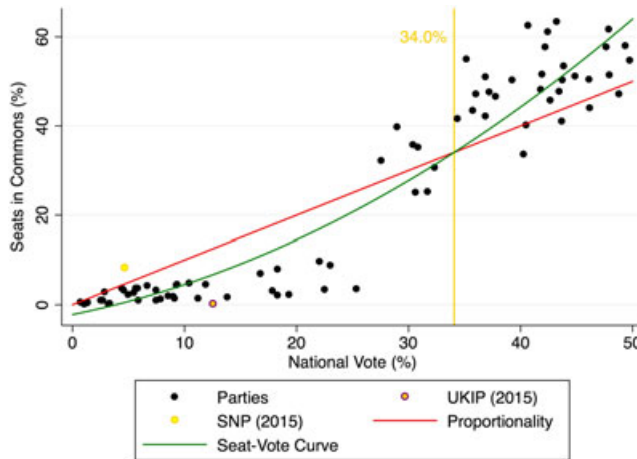


FIGURE 6 Votes and seats by party in the UK House of Commons: 1945–2015 (Sources: Richard, Kimber, Political Science Resources, www.politicsresources.net/area/uk/percentvote.htm; for 2015, www.bbc.com/news/election/2015/results). UKIP = United Kingdom Independence Party; SNP = Scottish National Party

however, for parties or minority groups that have significant geographic concentrations and that form local pluralities in one or more districts.

The graphs show several aspects of FPTP systems. First, the straight red line on each figure shows proportionality, or where parties would receive the same percentage of seats and votes earned, e.g., the 1952 U.S. election where Democratic candidates won 49.2% of the vote nationwide, and 49.1% of seats in the House (Figure 5). Such results would be typical in a proportional representation system, but are rare in FPTP ones. Values above the red line indicate elections where party candidates received a lower proportion of votes nationally than proportion of seats won. In the 2012 election, for example, the Republicans won 47% of the national vote, but obtained 54% of seats. Conversely, values below the line are where the party won fewer seats than their relative national vote share, such as the 1964 election when Republicans won over 42% of the national vote but only 32% of House seats.

Second, the green line on each figure shows a general “seat-vote curve” for each country’s elections during the postwar period. (This relationship is basically linear for the USA and curvilinear in the UK and Canada.) It shows

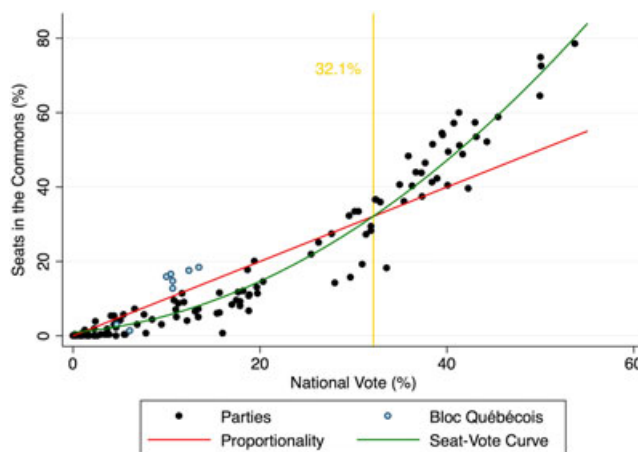


FIGURE 7 Votes and seats by party in the Canadian House of Commons: 1945–2015 (Source: Canadian Parliament, Library of Parliament, <https://lop.parl.ca/ParlInfo/Compilations/ElectionsAndRidings/ResultsParty.aspx>)

how many seats a party would be expected to win for a given share of the national vote. In all three cases, the seat-vote curve is lower than the line of proportionality for low vote totals, and higher after a threshold, marked by the vertical yellow line. (The thresholds are about 48% in the USA, 34% in the UK, and 32% in Canada.) In other words, the seat-vote relationship will tend to create a disadvantage for parties who earn votes below these thresholds, but creates an advantage for those who earn above it.

These seat-vote relationships hold in general, but the geographic distribution of a party's support has a profound effect (Johnston, Rossiter, Pattie, & Dorling, 2002). Typically, a party with geographically concentrated support will earn more seats relative to their national vote share. In 2015, for example, the Scottish National Party won over 8% of the seats in the House of Commons with less than 5% of the national vote in the UK election (Figure 6). Similarly, the Bloc Québécois party typically wins a higher share of seats than votes in Canada (Figure 7). In both cases, the parties' supporters are concentrated in one region of the country. This enables their candidates to win pluralities in constituencies in Scotland and Quebec, respectively, even though they win only a modest share of the national vote. Conversely (as shown in Figure 6), a party such as the United Kingdom Independence Party, whose support is spread among many constituencies may amass considerable support nationally (12.6%) while winning few seats (one out of 650 in 2015).

The fate of United Kingdom Independence Party in 2015 illustrates a general characteristic of FPTP systems: They can generate a large number of “wasted votes,” or votes cast for losing candidates (Johnston et al., 2002; Johnston, Rossiter, & Pattie, 2005). While this phenomenon occurs in most Congressional and Parliamentary elections, the winner-take-all U.S. Electoral College system means that it also effects presidential elections (Johnston et al., 2005). This occurred perhaps most dramatically in 2016 when Secretary Hillary Clinton lost despite winning a several-million popular vote majority with large margins in populous states like California and New York and narrow losses in Michigan, Pennsylvania, and Wisconsin.

From a party's perspective, the ideal set of constituencies is one that produces the most efficient distribution of their supporters, one that results in the fewest wasted votes for their candidates (Johnston et al., 2002). In theory, the most efficient distribution of votes is one where a party wins each victory by one vote, and thereby maximizes its seat-vote ratio. In practice, of course, parties do not attempt to cut their margins of victory so thinly.

The relationships among the general level of support for a party, the geographic distribution of support, the boundaries of electoral constituencies, and the way that parties and candidates respond to these conditions remain a central focus for electoral geography (Pattie & Johnston, 2003). Interest in this topic is also shared by some political scientists, who tend to emphasize comparative studies, including PR systems (e.g., Sellers, 2013).

5.2 | Redistricting and gerrymandering

Jurisdictions using plurality systems typically redraw their constituencies periodically in the process of redistricting (Morrill, 1981; Webster, 2000; Webster, 2013). Such exercises are often highly controversial because they provide opportunities for political parties to manipulate constituency boundaries to create gerrymanders (Martis, 2008; Monmonier, 2001; Taylor & Gudgin, 1976; Yoshinaka & Murphy, 2009). While there is no single measure of what constitutes gerrymandering, it is generally understood to mean a configuration of election district boundaries that confers an unfair advantage (or disadvantage) to a political party or minority group (Johnston, 2002a, 2002b; Rush, 2000; Webster, 2013). Although proportionality is an attractive baseline for measuring gerrymandering, the seat-vote ratio mean that even districting plans created through a non-partisan process will typically produce non-proportional election results. The determination of what constitutes an “unfair” advantage (or disadvantage) is ultimately a political decision rather than a scientific one (Forest, 2005).

There are three principal techniques of gerrymandering. The first is malapportionment, or the unequal distribution of electors among constituencies (Cox & Katz, 2002; Dixon, 1968; Johnston, 2002a, 2002b; Rowley, 1975). While the so-called “rotten borough” (a highly under populated constituency) was a long-standing British tradition, most countries now limit the degree of population deviation they permit among constituencies. The USA has among the strictest standards, which in practice require the equal distribution of population among Congressional districts within a particular state. Canada, in contrast, permits up to $\pm 25\%$ population deviation among ridings within a province as a matter of routine. One consequence is heavily populated urban ridings in Canada that dilute the voting strength of visible minorities as well as of city dwellers more generally (Forest, 2012). Like most countries, however, the distribution of seats among provinces, states, and other sub-national units means that some degree of malapportionment is probably unavoidable in FPTP systems (Norman, Purdam, Tajar, & Simpson, 2007).

Second, if a group is relatively small compared to the size of constituencies, it can be split among two or more districts (Monmonier, 2001). Such “cracking” works by lowering the number of a political or ethnic minority in a district below plurality, ensuring that they can never elect a representative of their choice (assuming voting is polarized among ethnic groups).

Finally, if a minority group is too large to be split effectively among constituency, it may be “packed” into one district to create a supermajority and thereby waste as many votes as possible. Methodologically and legally, the evaluation of packing can be difficult. The difference between a packed district and one with a “safe” majority is blurry, and depends on many factors, including the volatility of the electorate, turnout, citizenship rates, and population movement. Ethnicity and party may also intersect: In the USA, a district with a bare majority of African Americans may have a large Democratic majority. It would be competitive in racial terms, but packed from a partisan perspective.

Gerrymandering concerns the relationship between votes and seats, but is often conflated with the geographic shape of constituencies. That is, a district is often termed a gerrymander because it has a strange or “bizarre” shape, or its shape itself is taken as evidence that the boundaries have been “unfairly” manipulated (Forest, 2001). Congressional districts drawn in some U.S. states during the 1990s, such as the 1st and 12th districts in North Carolina, are often held up as prime examples of gerrymandering (Figure 8). The focus on shape is often misplaced, however, because a smooth boundary or compact shape is no guarantee against malapportionment, cracking, or packing.

Jurisdictions have grappled with the political nature of redistricting in various ways. Canada and the United Kingdom, for example, use arms-length boundary commissions in redistricting (Courtney, 2001; Rossiter, Pattie, & Johnston, 1999). Even such commissions, however, may be susceptible to partisan influence (Johnston, Pattie, & Rossiter, 1997; Rossiter, Johnston, & Pattie, 1998). In contrast, the redistricting process in the United States is typically under the jurisdiction of state legislatures and thus tends to be a highly partisan affair. The actions of legislatures are limited through the legal system, and litigation is a normal step in the U.S. redistricting process. For example, 31 of the 50 states had their redistricting plans challenged in court by the end of 2011 during that round

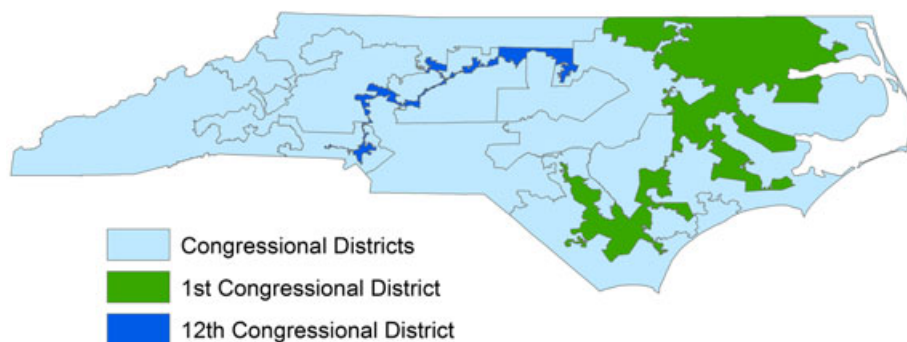


FIGURE 8 Congressional districts in North Carolina used for the 1992 election (103rd Congress). (Source: Map by author; data from U.S. Census Bureau www.census.gov/geo/maps-data/data/cbf/cbf_cds.html.)

of redistricting (Webster, 2013). Consequently, the electoral geography of redistricting in the USA tends to engage closely with Constitutional law (Forest, 2001; Forest, 2004). In such studies, the primary questions concern not whether a districting plan is biased, but how the court system judges the plan in Constitutional terms.

The centrality of racial conflict to the politics of representation also makes the study of redistricting in the U.S. distinct from other jurisdictions. Elsewhere, gerrymandering and bias are typically discussed only in terms of political parties, but in the USA, the representation of racial minorities, particularly African Americans and Latinos, is of at least equal concern (Forest, 2001; Leib, 1998; Lennertz, 2000).

6 | CONCLUSION

Early versions of electoral geography provided powerful tools to visualize the political character of states. Moreover, mapping votes was one of the best available ways to examine how social factors shape political preference. Systematic survey and polling data are generally superior for such causal inference, however, so that electoral geography lost favor for such questions. Geographers responded in several ways, including efforts to rigorously define spatial patterns and to analyze spatial variation in the causal relationships revealed by survey and polling data. This latter approach continues to provide a useful corrective to the idea that political relationships, dynamics, and institutions function uniformly over space. The ongoing challenge in this work, particularly in its conversation with political science, is to show not only that geographic context is significant, but to specify how much and in what ways it makes a difference. Electoral geography is arguably at its strongest when it steps back from its early focus on individual causal relationships ("why do voters have particular preferences?") to examine how political *institutions* use and are shaped by spatial relationships. Electoral systems (especially FPTP systems), political parties, and states themselves are organized geographically, and understanding this organization is essential to understanding their function. Moreover, political organizations are keenly aware of these spatial relationships and both shape their activities in response and seek to manipulate them through campaigns, legal action, public expenditures, infrastructure development, and the like. Recognizing this both broadens and deepens the scope of electoral geography.

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ENDNOTES

- ¹ In this article, I concentrate on works published in geography journals or written by academic geographers, and consequently, I only include a little of the substantial literatures in political science, political history, and related fields that address the geography of elections. Although this focus deemphasizes the interchange between such fields and electoral geography, my purpose is to characterize the development and state of electoral geography as a discipline. Students and scholars interested in the spatial dimension of politics should certainly avail themselves to the broader set of works in these other fields. Indeed, a basic search of a scholarly article database shows that the number of political science publications on “electoral systems” (nearly 3,500) exceeds the number in geography (just over 100) by an order of magnitude. The ratio is reversed, of course, for a topic such as “electoral cartography.” Finally, I have not included much from “critical” approaches to electoral geography that either consider electoral patterns as part of complex social analysis (e.g., Brown, Knopp, & Morrill, 2005; Nelson, 2006) or as an aspect of World System Theory (e.g., Flint, 2001).
- ² There is voluminous literature on the use of maps for nation-building (Anderson, 1983; Black, 1997; Branch, 2013; Harley & Laxton, 2001; Thongchai, 1994), but little attention specifically to the role of electoral cartography in this process.

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